

CHAHAT

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Portfolio: <https://chahat2830.github.io/-Portfolio-website/>

PROFESSIONAL SUMMARY

Computer Vision Engineer (Fresher) specialising in CNN-based deep learning architectures for real-world visual intelligence. Hands-on experience in building and evaluating **advanced computer vision pipelines** involving face detection, emotion recognition, and real-time inference, with a strong interest in solving **complex, real-world vision challenges** beyond academic baselines.

EDUCATION

Bachelor of Computer Applications (BCA) – Pursuing

Indira Gandhi National Open University (IGNOU)

Bachelor of Arts (B.A.) – Completed

Delhi University – School of Open Learning (DU SOL)

Senior Secondary (10+2)

Kendriya Vidyalaya

NIELIT Certifications

- 'O' Level (Information Technology)
 - 'A' Level (Advanced Information Technology)
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TECHNICAL SKILLS

Programming & Data

Python | SQL

Machine Learning & AI

Supervised & Unsupervised Learning | Feature Engineering | Model Evaluation

Deep Learning

Artificial Neural Networks (ANN) | Convolutional Neural Networks (CNN) | Transfer Learning | Model Training

Computer Vision

OpenCV | Image Preprocessing | Face Detection | Facial Expression Recognition | Image Classification

Tools & Platforms

Git | Jupyter Notebook | VS Code

PROJECTS

Advanced Project

Real-Time Facial Emotion Recognition System (CNN-Based)

- Designed a real-time facial emotion recognition pipeline using **CNN architectures**
- Implemented **face detection and preprocessing** using OpenCV
- Classified facial expressions and performed **real-time inference**
- Demonstrated practical deployment of **computer vision + deep learning models**
- Classified facial expressions with high class-balanced performance in real-time inference

Additional Projects

Book Recommender System

- Implemented content-based and popularity-based recommendation techniques
- Applied similarity measures for personalized recommendations

Olympics Analytics Dashboard

- Developed a data visualization dashboard to analyze Olympic performance trends

COVID-19 Pandemic Dashboard

- Visualized global COVID-19 statistics using structured datasets

SMS Spam Detection System

- Built a machine learning classifier for spam detection
- Applied text preprocessing and classification algorithms

RESEARCH EXPERIENCE

Explainable Imbalance-Aware Facial Emotion Recognition

Manuscript in Preparation

Role: First Author

Affiliation: National Institute of Electronics and Information Technology (NIELIT), Delhi

- Designed an **imbalance-aware facial emotion recognition (FER) framework** using **ConvNeXt V2-Tiny** for real-world affective computing applications.
- Initialized the backbone with **self-supervised FCMAE pretraining (ImageNet-22K & ImageNet-1K)** to improve feature generalization under limited data.
- Addressed severe class imbalance using **Class-Weighted Focal Loss**, significantly improving minority emotion recognition (Fear class F1-score: **0.91**).
- Integrated **Grad-CAM–based explainability** to validate that model predictions rely on semantically meaningful facial regions.
- Achieved **macro-F1 score of 0.94**, outperforming strong CNN baselines including DenseNet121 and EfficientNetV2

EXPERIENCE

Project Assistant / Teaching Support (Part-Time) NIELIT

- Assisted in academic projects related to IT and computer fundamentals
 - Supported teaching activities and guided learners during practical sessions
 - Helped students with Python programming and introductory machine learning concepts
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CERTIFICATIONS

- Information Technology
- Web Development
- Python Programming
- Internet of Things (IoT)
- Cloud Computing
- Artificial Intelligence Bootcamp

In Progress:

- Data Science
 - Database Systems
 - Artificial Intelligence
 - Machine Learning
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INTERESTS

Computer Vision | Explainable AI (XAI) | Real-Time Vision Systems | Intelligent Automation